Math 54 — Week 12

Nonhomogeneous equations: undetermined coefficients and superposition (Nagle 4.4–4.5)

§1 Facts used from Weeks 1–11

Throughout, $a \neq 0$, b, c are real constants, $L[y] = ay'' + by' + cy$, and $p(r) = ar^2 + br + c$ is the auxiliary polynomial.

(W1) **Homogeneous equations** (Week 11, Nagle 4.2–4.3). L is linear: $L[c_1y_1 + c_2y_2] = c_1L[y_1] + c_2L[y_2]$. $L[e^{rt}] = p(r)e^{rt}$. The problem $L[y] = 0$, $y(t_0) = Y_0$, $y'(t_0) = Y_1$ has exactly one solution (Nagle Thm. 1). Let y_1, y_2 solve $L[y] = 0$, with Wronskian $W = y_1y_2' - y_1'y_2$. If $W(\tau) = 0$ at one point τ, then y_1, y_2 are linearly dependent (Nagle Lemma 1). So two independent solutions have $W(t) \neq 0$ at every t, and then every solution is $c_1y_1 + c_2y_2$ (Nagle Thm. 2).

(W2) **The three root cases** (Week 11). Distinct real roots r_1, r_2: e^{r_1t}, e^{r_2t}. A double root r: e^{rt}, te^{rt}, and then $2ar + b = 0$, that is, $p'(r) = 0$. Complex roots $\alpha \pm i\beta$ ($\beta \neq 0$): $e^{\alpha t}\cos\beta t$, $e^{\alpha t}\sin\beta t$.

(W3) **Complex exponentials** (Week 11). $e^{i\theta} = \cos\theta + i\sin\theta$; $\frac{d}{dt}e^{rt} = re^{rt}$ also for complex r. If $z = u + iv$ is complex-valued and a, b, c are real, then $L[z] = L[u] + iL[v]$ with $L[u]$, $L[v]$ real (the computation in Nagle Lemma 2), so $L[\operatorname{Re} z] = \operatorname{Re} L[z]$ and $L[\operatorname{Im} z] = \operatorname{Im} L[z]$.

(W4) **Triangular matrices** (Weeks 4–5). The determinant of a triangular matrix is the product of its diagonal entries, so a triangular matrix with nonzero diagonal is invertible.

(W5) **Solutions of $A\mathbf{x} = \mathbf{b}$** (Week 2, Lay 1.5 Thm. 6). If $A\mathbf{p} = \mathbf{b}$, the solutions of $A\mathbf{x} = \mathbf{b}$ are exactly $\mathbf{p} + \mathbf{v}_h$ with $A\mathbf{v}_h = \mathbf{0}$.

(W6) **The forced mass–spring model** (Nagle 4.1, cited). A mass m on a spring of stiffness k with damping b, pushed by an outside force $F(t)$, has position y with $my'' + by' + ky = F(t)$.

§2 This week

§2.1 The method of undetermined coefficients (Nagle 4.4)

The question. Find one function y with $y'' + 3y' + 2y = 3t$. The guess $y = At$ looks natural, since the right side is a multiple of t. Why does it fail, and what guess works?

An equation $L[y] = f(t)$ with $f \neq 0$ is **nonhomogeneous**; one solution of it is a **particular solution** y_p. This section finds y_p when f is a polynomial, an exponential, a sine or cosine, or a product of these. The idea: choose a family of functions that L maps into itself and that contains f, write y_p in that family with unknown coefficients, and match coefficients.

Example 1 (Nagle): a polynomial right side. For $y'' + 3y' + 2y = 3t$, try $y = At$: then $y' = A$, $y'' = 0$, and $L[y] = 3A + 2At$. Matching $3t$ needs $2A = 3$ and $3A = 0$, which is impossible. The derivative of At produced a constant term that the guess cannot cancel. So add a constant: $y = At + B$ gives

$$y'' + 3y' + 2y = 3A + 2(At + B) = 2At + (3A + 2B).$$

Matching $3t$: $2A = 3$ and $3A + 2B = 0$, so $A = \frac{3}{2}$, $B = -\frac{9}{4}$, and $y_p = \frac{3}{2}t - \frac{9}{4}$.

The question answered. At fails because L sends t to $2t+3$: derivatives lower the degree, so lower powers appear. The guess must contain every power up to the degree of f: $y_p = At + B$, giving $y_p = \frac{3}{2}t - \frac{9}{4}$.

Examples 2 and 3 (Nagle): exponential and sine. **(2)** $y'' + 3y' + 2y = 10e^{3t}$. Try Ae^{3t}: by (W1),

$$L[Ae^{3t}] = Ap(3)e^{3t} = A(9+9+2)e^{3t} = 20Ae^{3t},$$

so $A = \frac{1}{2}$ and $y_p = \frac{1}{2}e^{3t}$.

(3) $y'' + 3y' + 2y = \sin t$. The guess $A\sin t$ fails, since its derivative is a cosine: $L[A\sin t] = A\sin t + 3A\cos t$. Include both: with $y = A\sin t + B\cos t$, $y' = A\cos t - B\sin t$, $y'' = -A\sin t - B\cos t$,

$$y'' + 3y' + 2y = (A - 3B)\sin t + (3A + B)\cos t.$$

Matching $\sin t$: $A - 3B = 1$, $3A + B = 0$, so $A = \frac{1}{10}$, $B = -\frac{3}{10}$, and $y_p = \frac{1}{10}\sin t - \frac{3}{10}\cos t$.

Physically, a damped spring (W6) shaken by $\sin t$ ends up moving at the same frequency (Nagle 4.1, cited), so a combination of $\cos t$ and $\sin t$ is the expected response.

Example 4 (Nagle): polynomial times exponential. $y'' + 4y = 5t^2e^t$. Try $y = (At^2 + Bt + C)e^t$. Then

$$y' = (At^2 + (2A+B)t + B + C)e^t, \qquad y'' = (At^2 + (4A+B)t + 2A + 2B + C)e^t,$$

$$y'' + 4y = \big(5At^2 + (4A+5B)t + (2A+2B+5C)\big)e^t.$$

Matching $5t^2e^t$: $5A = 5$, $4A + 5B = 0$, $2A + 2B + 5C = 0$, so $A = 1$, $B = -\frac{4}{5}$, $C = -\frac{2}{25}$, and

$$y_p = \Big(t^2 - \frac{4}{5}t - \frac{2}{25}\Big)e^t.$$

When the basic guess fails. For $y'' + y' = 5$, the guess $y = A$ gives $L[A] = 0$: the constant solves the homogeneous equation, so no choice of A gives 5. Multiplying by t works: $L[At] = 0 + A = A$, so $A = 5$ and $y_p = 5t$. For $y'' - 6y' + 9y = e^{3t}$, both Ae^{3t} and Ate^{3t} solve the homogeneous equation (3 is a double root), and one needs At^2e^{3t}:

$$L[At^2e^{3t}] = A(2 + 12t + 9t^2)e^{3t} - 6A(2t + 3t^2)e^{3t} + 9At^2e^{3t} = 2Ae^{3t},$$

so $A = \frac{1}{2}$ and $y_p = \frac{1}{2}t^2e^{3t}$. The number of extra factors of t is the number of times the exponent is a root of p. The next formula shows why.

Proposition (The key formula)

For a polynomial $q(t)$ and a real or complex number r,

$$L\big[q(t)e^{rt}\big] = \big(a\,q'' + p'(r)\,q' + p(r)\,q\big)e^{rt}, \qquad p'(r) = 2ar + b.$$

Proof. By the product rule (and (W3) for complex r), $(qe^{rt})' = (q' + rq)e^{rt}$ and $(qe^{rt})'' = (q'' + 2rq' + r^2q)e^{rt}$. So

$$L[qe^{rt}] = \big(a(q'' + 2rq' + r^2q) + b(q' + rq) + cq\big)e^{rt} = \big(aq'' + (2ar+b)q' + (ar^2 + br + c)q\big)e^{rt}. \qquad \square$$

So matching $L[qe^{rt}] = Ct^me^{rt}$ means solving $aq'' + p'(r)q' + p(r)q = Ct^m$ for a polynomial q. Which coefficient leads depends on whether $p(r)$ and $p'(r)$ are 0.

Where the rule comes from (Nagle). Write $q = A_nt^n + A_{n-1}t^{n-1} + A_{n-2}t^{n-2} + \dots$ with $A_n \neq 0$. By the key formula, the three highest powers in $L[qe^{rt}]$ are

$$\begin{aligned}
&A_n\,p(r)\,t^ne^{rt},\\
&\big(nA_n\,p'(r) + A_{n-1}\,p(r)\big)t^{n-1}e^{rt},\\
&\big(n(n-1)A_n\,a + (n-1)A_{n-1}\,p'(r) + A_{n-2}\,p(r)\big)t^{n-2}e^{rt}.
\end{aligned}$$

To match Ct^me^{rt}, the highest power present must be t^m. If $p(r) \neq 0$, the first line is the highest, so $n = m$. If $p(r) = 0 \neq p'(r)$, the first line is 0 and the second is the highest, so $n = m + 1$. If $p(r) = p'(r) = 0$, only the third

remains, so $n = m+2$. In the last two cases the lowest terms $A_0 e^{rt}$ (and $A_1 te^{rt}$) solve the homogeneous equation and can be set to 0, which leaves a factor t (or t^2). The case $f = Ct^m$ is $r = 0$: then one factor t is needed if $c = 0 \neq b$, and two if $b = c = 0$.

Theorem (Method of undetermined coefficients (Nagle 4.4))

(a) For $L[y] = Ct^m e^{rt}$ (r real, $m \geq 0$ an integer) there is a particular solution

$$y_p = t^s(A_m t^m + \cdots + A_1 t + A_0)e^{rt},$$

where $s = 0$ if r is not a root of p, $s = 1$ if r is a **simple root** ($p(t) = a(t-r)(t-r_2)$ with $r_2 \neq r$), and $s = 2$ if r is a double root.

(b) For $L[y] = Ct^m e^{\alpha t}\cos\beta t$ or $Ct^m e^{\alpha t}\sin\beta t$ with $\beta \neq 0$ there is a particular solution

$$y_p = t^s(A_m t^m + \cdots + A_0)e^{\alpha t}\cos\beta t + t^s(B_m t^m + \cdots + B_0)e^{\alpha t}\sin\beta t,$$

where $s = 0$ if $\alpha + i\beta$ is not a root of p and $s = 1$ if it is.

The idea

By the key formula, the problem is to find a polynomial q with $aq'' + p'(r)q' + p(r)q = Ct^m$. If $p(r) \neq 0$, the map $q \mapsto aq'' + p'(r)q' + p(r)q$ keeps the degree of q and is invertible on polynomials of degree $\leq m$, because its matrix is triangular with $p(r)$ on the diagonal. If $p(r) = 0$, the term $p(r)q$ is 0 and the left side has degree one less than q, so q needs degree $m+1$; its constant term solves the homogeneous equation and can be dropped, which leaves a factor t. At a double root $p'(r)q'$ is 0 too, which requires a second factor t.

Proof. Write $\mathbb{P}_m$ for the polynomials of degree at most m, with basis $1, t, \ldots, t^m$, and $M(q) = aq'' + p'(r)q' + p(r)q$.

1. **r is not a root ($s = 0$).** $M(t^k) = p(r)t^k + (\text{lower powers})$, because q' and q'' have lower degree. So M maps $\mathbb{P}_m$ to $\mathbb{P}_m$, and its matrix in the basis $1, t, \ldots, t^m$ is upper triangular with every diagonal entry $p(r) \neq 0$. By (W4) it is invertible, so some $q \in \mathbb{P}_m$ has $M(q) = Ct^m$.

2. **r is a simple root ($s = 1$).** Now $p(r) = 0$, and $p'(r) \neq 0$: from $p(t) = a(t-r)(t-r_2)$, $p'(r) = a(r-r_2) \neq 0$. So

   $$M(q) = aq'' + p'(r)q' = N(q'), \qquad N(u) = au' + p'(r)u.$$

   As in step 1, N is invertible on $\mathbb{P}_m$ (diagonal entries $p'(r) \neq 0$), so some $u \in \mathbb{P}_m$ has $N(u) = Ct^m$. Take $q(t) = \int_0^t u$: then $q' = u$, $M(q) = Ct^m$, and $q(0) = 0$, so $q = t \cdot (\text{polynomial of degree } m)$.

3. **r is a double root ($s = 2$).** Now $p(r) = p'(r) = 0$ (W2), so $M(q) = aq''$, and the polynomial below has $aq'' = Ct^m$:

   $$q = \frac{C}{a(m+1)(m+2)}\, t^{m+2}.$$

   This q is t^2 times a polynomial of degree m.

4. **Part (b): complex exponent $r = \alpha + i\beta$.** The key formula holds for complex r, and M now has complex coefficients. Steps 1 and 2 used only that a triangular matrix with nonzero diagonal is invertible (W4), whose proof is the same for complex entries. If $p(r) \neq 0$, step 1 applies with diagonal entries $p(r)$. If $p(r) = 0$, then r is a simple root: the roots $\alpha \pm i\beta$ differ because $\beta \neq 0$, and since their sum 2α equals $-b/a$, we get $b = -2a\alpha$ and $p'(r) = 2a(\alpha + i\beta) + b = 2ai\beta \neq 0$; so step 2 applies with diagonal entries $p'(r)$. Either way $s \leq 1$, and there is a complex polynomial $Q = Q_1 + iQ_2$ (Q_1, Q_2 real, of degree at most m) with

   $$L\left[t^s Q e^{rt}\right] = Ct^m e^{rt}.$$

5. Since $Ct^m e^{rt} = Ct^m e^{\alpha t}(\cos\beta t + i\sin\beta t)$, taking real and imaginary parts with (W3) gives solutions for the cosine and the sine right sides:

   $$\mathrm{Re}\left(t^s Q e^{rt}\right) = t^s e^{\alpha t}(Q_1\cos\beta t - Q_2\sin\beta t), \qquad \mathrm{Im}\left(t^s Q e^{rt}\right) = t^s e^{\alpha t}(Q_2\cos\beta t + Q_1\sin\beta t).$$

   Both have the form in (b).

□

First-order equations $by' + cy = f$ $(b \neq 0)$. Set $a = 0$: the key formula becomes $L[qe^{rt}] = \big(bq' + p(r)q\big)e^{rt}$ with $p(r) = br + c$ and $p'(r) = b \neq 0$. Steps 1 and 2 of the proof apply, so $s = 0$ unless $r = -c/b$, and then $s = 1$. Instance: for $y' + 2y = 4t$, $y = At + B$ gives $2At + (A + 2B)$, so $A = 2$, $B = -1$ and $y_p = 2t - 1$.

Examples 5 and 6 (Nagle): writing down the form. For $y'' + 2y' - 3y = f$, $p(r) = (r-1)(r+3)$. For $y'' - 2y' + y = f$, $p(r) = (r-1)^2$.

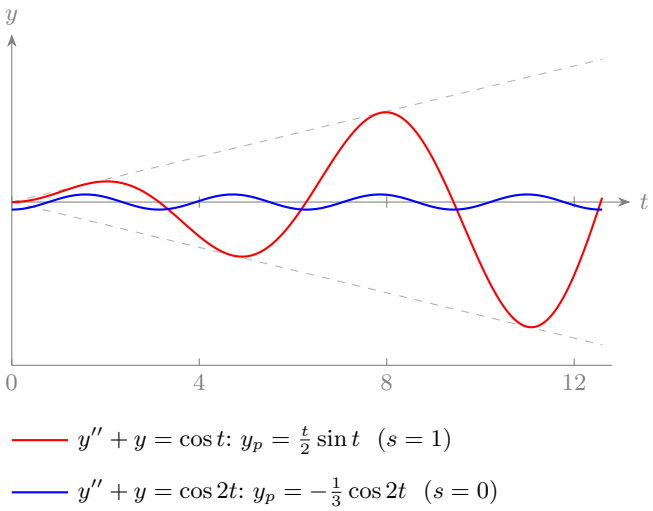
f	$y'' + 2y' - 3y = f$	$y'' - 2y' + y = f$
$7\cos 3t$	$A\cos 3t + B\sin 3t$	same
$2te^t \sin t$	$(A_1t + A_0)e^t\cos t + (B_1t + B_0)e^t \sin t$	same
$t^2 \cos \pi t$	$(A_2t^2 + A_1t + A_0)\cos \pi t + (B_2t^2 + B_1t + B_0)\sin \pi t$	same
$5e^{-3t}$	Ate^{-3t} $\quad(-3$ simple root$)$	Ae^{-3t}
$3te^t$	$t(A_1t + A_0)e^t$ $\quad(1$ simple$)$	$t^2(A_1t + A_0)e^t$ $\quad(1$ double$)$
t^2e^t	$t(A_2t^2 + A_1t + A_0)e^t$	$t^2(A_2t^2 + A_1t + A_0)e^t$

Example 7 (Nagle): a complex root. For $y'' - 2y' + 2y = 5te^t \cos t$, $p(r) = r^2 - 2r + 2$ has roots $1 \pm i$, and $\alpha + i\beta = 1 + i$ is one of them, so $s = 1$:

$$y_p = t(A_1t + A_0)e^t \cos t + t(B_1t + B_0)e^t \sin t.$$

Resonance (Nagle 4.10). When the forcing has the same frequency as a homogeneous solution, $s = 1$ and the amplitude of y_p grows without bound. For $y'' + y = \cos t$, $p(r) = r^2 + 1$ has the root i, and $p'(i) = 2i \neq 0$. The key formula with $q = t$ and $r = i$ gives $L[te^{it}] = p'(i)e^{it} = 2ie^{it}$, so $z = te^{it}/(2i)$ solves $L[z] = e^{it}$, and by (W3) $y_p = \operatorname{Re} z$:

$$z = \frac{t(\cos t + i \sin t)}{2i} = \frac{t}{2}\sin t - \frac{it}{2}\cos t, \qquad y_p = \operatorname{Re} z = \frac{t}{2}\sin t.$$



$p(r) = r^2 + 1$ has roots $\pm i$. For $\cos t$, i is a root, so $s = 1$ and the particular solution grows inside the dashed lines $\pm t/2$ (resonance: the forcing has the frequency of the homogeneous solutions $\cos t$, $\sin t$). For $\cos 2t$, $2i$ is not a root: $s = 0$ and the response stays bounded.

The method needs right sides whose derivatives stay in a finite family. For $\tan t$ (or $\sec t$, $1/t$, $t^{-1}e^t$) the derivatives keep producing new functions, so no finite guess works; Nagle 4.6 treats these by another method. It also needs constant coefficients: $ty'' - y' + 2y = \sin 3t$ is not covered.

Shortcuts

- **Exponential right side**: if $p(r) \neq 0$, then $y_p = \big(C/p(r)\big)e^{rt}$ for $f = Ce^{rt}$. Reason: $L[e^{rt}] = p(r)e^{rt}$ (W1). Instance: Example 2, $y_p = \frac{10}{20}e^{3t}$.

- **Sine or cosine by complex exponential**: for $f = C\sin\beta t$, solve $L[z] = Ce^{i\beta t}$ by $z = \big(C/p(i\beta)\big)e^{i\beta t}$ and take $y_p = \operatorname{Im} z$ (for cos, $\operatorname{Re} z$). Reason: (W3). Condition: $p(i\beta) \neq 0$. Instance: Example 3, $p(i) = 1 + 3i$, and
$$z = \frac{e^{it}}{1+3i} = \frac{(1-3i)(\cos t + i\sin t)}{10}, \qquad \operatorname{Im} z = \frac{\sin t - 3\cos t}{10}.$$
- **Resonant sine or cosine**: if $p(i\beta) = 0 \neq p'(i\beta)$, then $z = \big(C/p'(i\beta)\big)te^{i\beta t}$ solves $L[z] = Ce^{i\beta t}$; take $\operatorname{Re} z$ or $\operatorname{Im} z$. Reason: the key formula with $q = t$ gives $L[te^{rt}] = p'(r)e^{rt}$ for complex r too. Instance: $y'' + y = \cos t$ gives $y_p = \frac{t}{2}\sin t$ (computed above).
- **Finding** s: compute the roots of p once; then s is how many times the exponent (r, or $\alpha + i\beta$) appears among them. Reason: the Theorem. Instance: the table above.
- **Double root shortcut**: if r is a double root, $L[qe^{rt}] = aq''e^{rt}$, so integrate twice. Reason: the key formula with $p(r) = p'(r) = 0$. Instances: $y'' - 6y' + 9y = e^{3t}$ gives $q'' = 1$, $q = \frac{1}{2}t^2$; and $y'' = 2$ (0 is a double root of r^2) gives $q = t^2$, which is two factors of t times the homogeneous solution 1.
- **Exponent is a simple root**: if $p(r) = 0 \neq p'(r)$, then $y_p = \big(C/p'(r)\big)te^{rt}$ for $f = Ce^{rt}$. Reason: the key formula with $q = t$ gives $L[te^{rt}] = p'(r)e^{rt}$. Instance: $y'' + y' = 5$: $r = 0$, $p'(0) = 1$, $y_p = 5t$.

Traps

- Missing lower powers: for $y'' + 3y' + 2y = 3t$ the guess At fails (Example 1); keep all of $t^m, \ldots, t, 1$.
- Missing the partner trig function: for $\sin t$ the guess $A\sin t$ fails (Example 3); always include both $\cos\beta t$ and $\sin\beta t$.
- Forgetting t^s: for $y'' + y' = 5$, the guess A gives $0 = 5$. Check the roots of p before guessing.
- Too few factors of t fail, and so do too many. For $y'' + 3y' + 2y = 10e^{3t}$, the guess Ate^{3t} gives $L[Ate^{3t}] = A(20t + 9)e^{3t}$, which never equals $10e^{3t}$. Use exactly t^s. A complex root is always simple, so $s \leq 1$ in part (b): never t^2 with cos and sin for a second-order equation.

§2.2 Superposition and the general solution (Nagle 4.5)

The question. We know $y_1 = \frac{3}{2}t - \frac{9}{4}$ solves $y'' + 3y' + 2y = 3t$ and $y_2 = \frac{1}{2}e^{3t}$ solves $y'' + 3y' + 2y = 10e^{3t}$. Find a solution of $y'' + 3y' + 2y = -9t + 20e^{3t}$ without new computations, and then all solutions.

Theorem (Superposition principle (Nagle Thm. 3))
If $L[y_1] = f_1$ and $L[y_2] = f_2$, then $L[k_1y_1 + k_2y_2] = k_1f_1 + k_2f_2$ for all constants k_1, k_2.

Proof. L is linear (W1), so
$$L[k_1y_1 + k_2y_2] = k_1L[y_1] + k_2L[y_2] = k_1f_1 + k_2f_2.$$ □

Instance (Nagle 4.5, Example 1): $y_1 + y_2 = \frac{3}{2}t - \frac{9}{4} + \frac{1}{2}e^{3t}$ solves $y'' + 3y' + 2y = 3t + 10e^{3t}$.

The question answered (first part). $-9t + 20e^{3t} = -3(3t) + 2(10e^{3t})$, so $-3y_1 + 2y_2 = -\frac{9}{2}t + \frac{27}{4} + e^{3t}$ is a solution.

Theorem (Existence and uniqueness, nonhomogeneous case (Nagle Thm. 4))

Let y_p solve $L[y] = f$ on an interval I containing t_0, and let y_1, y_2 be linearly independent solutions of $L[y] = 0$. Then for all Y_0, Y_1 the problem

$$L[y] = f, \qquad y(t_0) = Y_0, \quad y'(t_0) = Y_1$$

has exactly one solution on I, and it is $y = y_p + c_1 y_1 + c_2 y_2$ for suitable constants c_1, c_2.

The idea

The difference of two solutions of $L[y] = f$ solves $L[y] = 0$, exactly as the difference of two solutions of $A\mathbf{x} = \mathbf{b}$ solves $A\mathbf{x} = \mathbf{0}$ (W5). So all the freedom is in the homogeneous part, whose two constants are fixed by two conditions; the Wronskian is the determinant of that 2×2 system.

Proof. 1. **It is a solution.** By superposition, $L[y_p + c_1 y_1 + c_2 y_2] = f + 0 + 0 = f$.

2. **The conditions can be met.** They read, as a linear system in c_1, c_2,

$$c_1 y_1(t_0) + c_2 y_2(t_0) = Y_0 - y_p(t_0), \qquad c_1 y_1'(t_0) + c_2 y_2'(t_0) = Y_1 - y_p'(t_0).$$

Its determinant is the Wronskian $W(t_0)$, which is not 0 because y_1, y_2 are independent solutions (W1). So it has exactly one solution (c_1, c_2); by Cramer's rule,

$$c_1 = \frac{(Y_0 - y_p(t_0))y_2'(t_0) - (Y_1 - y_p'(t_0))y_2(t_0)}{W(t_0)}, \qquad c_2 = \frac{(Y_1 - y_p'(t_0))y_1(t_0) - (Y_0 - y_p(t_0))y_1'(t_0)}{W(t_0)}.$$

3. **Uniqueness.** If $\tilde{y}$ is another solution of the problem, $u = y - \tilde{y}$ has $L[u] = f - f = 0$ and $u(t_0) = u'(t_0) = 0$. The zero function also solves this, so by Nagle Thm. 1 (W1), $u = 0$.

□

So $y = y_p + c_1 y_1 + c_2 y_2$ is the **general solution** of $L[y] = f$: any solution y_g equals it, by choosing c_1, c_2 to match $y_g(t_0)$, $y_g'(t_0)$ and using uniqueness.

The question answered (second part). $p(r) = r^2 + 3r + 2 = (r+1)(r+2)$, so all solutions are

$$y = -\frac{9}{2}t + \frac{27}{4} + e^{3t} + c_1 e^{-t} + c_2 e^{-2t}.$$

Solving $L[y] = f$ with initial conditions.

1. Solve $p(r) = 0$; write the homogeneous solution $c_1 y_1 + c_2 y_2$ (W2).
2. Split f into terms of the forms in the theorem; find y_p for each (one guess can cover several terms with the same exponent), and add (superposition).
3. Apply the initial conditions to the *whole* $y = y_p + c_1 y_1 + c_2 y_2$ and solve for c_1, c_2.

Example 2 (Nagle): initial conditions. $y_p = t^2$ solves $y'' - y = 2 - t^2$: indeed $y_p'' - y_p = 2 - t^2$. The roots of $r^2 - 1$ are ± 1, so $y = t^2 + c_1 e^t + c_2 e^{-t}$. Then $y(0) = c_1 + c_2 = 1$ and $y'(0) = 0 + c_1 - c_2 = 0$ give $c_1 = c_2 = \frac{1}{2}$:

$$y = t^2 + \tfrac{1}{2}(e^t + e^{-t}) = t^2 + \cosh t.$$

Example 3 (Nagle): a forced, damped spring. Mass 1, damping 2, stiffness 2, force $5\sin t + 5\cos t$ (W6), with $y(0) = 1$, $y'(0) = 2$:

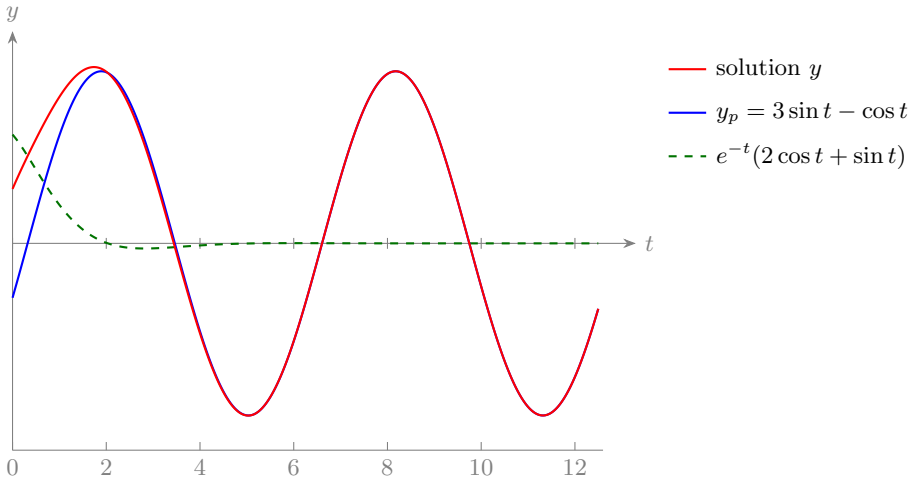
$$y'' + 2y' + 2y = 5\sin t + 5\cos t.$$

The roots of $r^2 + 2r + 2$ are $-1 \pm i$, so the homogeneous solution is $e^{-t}(c_1 \cos t + c_2 \sin t)$. Since i is not a root, $A\sin t + B\cos t$ is the form for each term. Nagle first solves each one: for $5\sin t$ the guess gives $\sin t - 2\cos t$, for $5\cos t$ it gives $2\sin t + \cos t$, and superposition adds them. The same form also handles both terms at once:

$$y_p'' + 2y_p' + 2y_p = (A - 2B)\sin t + (2A + B)\cos t,$$

and $A-2B=5$, $2A+B=5$ give $A=3$, $B=-1$. With $y=e^{-t}(c_1\cos t+c_2\sin t)+3\sin t-\cos t$, the conditions are $y(0)=c_1-1=1$ and $y'(0)=-c_1+c_2+3=2$, so $c_1=2$, $c_2=1$:

$$y=2e^{-t}\cos t+e^{-t}\sin t+3\sin t-\cos t.$$



Example 3. The homogeneous part (dashed) decays like e^{-t} (the *transient*); after a few seconds the solution (red) follows the particular solution (blue), which oscillates with the frequency of the force (the *steady state*).

Example 4 (Nagle): one guess for two terms. $y''-y=8te^t+2e^t$. The roots of r^2-1 are ±1, so 1 is a simple root and both terms need $s=1$: $8te^t$ needs $t(A_1t+A_0)e^t$, and $2e^t$ needs tA_0e^t, which is already contained in the first form. With $y_p=(A_1t^2+A_0t)e^t$, the key formula ($q=A_1t^2+A_0t$, $p(1)=0$, $p'(1)=2$, $a=1$) gives

$$L[y_p]=\big(2A_1+2(2A_1t+A_0)\big)e^t=\big(4A_1t+(2A_1+2A_0)\big)e^t.$$

Matching $8te^t+2e^t$: $A_1=2$, $A_0=-1$, so $y_p=(2t^2-t)e^t$.

Theorem (Method of undetermined coefficients, revisited (Nagle 4.5))

For $L[y]=P_m(t)e^{rt}$ with P_m a polynomial of degree m, use $y_p=t^s(A_mt^m+\cdots+A_0)e^{rt}$ with s as before. For $L[y]=P_m(t)e^{\alpha t}\cos\beta t+Q_n(t)e^{\alpha t}\sin\beta t$ ($\beta\neq0$), use

$$y_p=t^s(A_kt^k+\cdots+A_0)e^{\alpha t}\cos\beta t+t^s(B_kt^k+\cdots+B_0)e^{\alpha t}\sin\beta t,\qquad k=\max(m,n),$$

with $s=1$ if $\alpha+i\beta$ is a root of p and $s=0$ otherwise.

Proof. 1. **Exponential case.** Write $P_m=C_mt^m+\cdots+C_0$. Each term $C_jt^je^{rt}$ ($j\le m$) has a particular solution t^s(polynomial of degree j)e^{rt} by part (a). By superposition their sum solves $L[y]=P_m(t)e^{rt}$, and it is t^s times a polynomial of degree at most m times e^{rt}: the stated form.

2. **Trigonometric case.** Each term $t^je^{\alpha t}\cos\beta t$ or $t^je^{\alpha t}\sin\beta t$ with $j\le k$ has a particular solution of the form in part (b) with degree j. That form is contained in the form with degree k (set the higher coefficients to 0), so by superposition the sum has the stated form.

□

Examples 5 and 6 (Nagle): forms only. **(5)** $y''+2y'+2y=5e^{-t}\sin t+5t^3e^{-t}\cos t$. The roots are $-1\pm i$ and $\alpha+i\beta=-1+i$ is one of them, so $s=1$, and $k=\max(0,3)=3$:

$$y_p=t(A_3t^3+A_2t^2+A_1t+A_0)e^{-t}\cos t+t(B_3t^3+B_2t^2+B_1t+B_0)e^{-t}\sin t.$$

(6) Higher order works the same way (Nagle Ch. 6, cited: s is the number of times the exponent is a root of p): $y'''+2y''+y'=5e^{-t}\sin t+3+7te^{-t}$ has $p(r)=r(r+1)^2$, with 0 simple and -1 double. Term by term

the forms are $Ae^{-t}\cos t + Be^{-t}\sin t$, then Ct, then $t^2(Dt+E)e^{-t}$. (If -1 were a triple root, $7te^{-t}$ would need $t^3(Dt+E)e^{-t}$.) So

$$y_p = Ae^{-t}\cos t + Be^{-t}\sin t + Ct + t^2(Dt+E)e^{-t}.$$

Shortcuts

- **Reuse known particular solutions**: write the new right side as a combination of old ones and combine their solutions (superposition). Instance: the question.
- **One guess for terms with the same exponent**: terms $t^j e^{rt}$ with the same r share one form with the largest power. Reason: the form for $t^j e^{rt}$ with $j \le m$ is contained in the form for $t^m e^{rt}$. Instance: Example 4.
- **Powers of sines and cosines**: rewrite them first, then the method applies. Reason: $\cos 2t = 1 - 2\sin^2 t = 2\cos^2 t - 1$ and $\cos 3t = 4\cos^3 t - 3\cos t$, so $\sin^2 t = \frac{1}{2} - \frac{1}{2}\cos 2t$ and $\cos^3 t = \frac{3}{4}\cos t + \frac{1}{4}\cos 3t$; also $4x\sin^2 x + 4x\cos^2 x = 4x$. Instance: for $y'' + y = \sin^2 t$, the term $\frac{1}{2}$ gives $\frac{1}{2}$, and $-\frac{1}{2}\cos 2t$ gives $\frac{-1/2}{p(2i)}\cos 2t = \frac{1}{6}\cos 2t$, so $y_p = \frac{1}{2} + \frac{1}{6}\cos 2t$.
- **Hyperbolic functions**: $\cosh t = \frac{1}{2}(e^t + e^{-t})$ and $\sinh t = \frac{1}{2}(e^t - e^{-t})$ are two exponential terms. Instance: for $y'' - 4y = \cosh t$, $p(\pm 1) = -3$, so $y_p = -\frac{1}{6}e^t - \frac{1}{6}e^{-t} = -\frac{1}{3}\cosh t$.
- **Check**: substitute y_p back; and $y(t_0)$, $y'(t_0)$ of the final answer must match the data. Instance: in Example 3, $y(0) = 2 - 1 = 1$.

Traps

- Initial conditions go on the whole solution $y_p + c_1 y_1 + c_2 y_2$, not on the homogeneous part alone. In Example 2, solving $c_1 + c_2 = 1$, $c_1 - c_2 = 0$ happens to agree only because $y_p(0) = y_p'(0) = 0$; in Example 3, ignoring y_p would give $c_1 = 1$, $c_2 = 3$ instead of 2, 1.
- Superposition needs the same left side L. $y_1 = 1$ solves $y'' + y = 1$ and $y_2 = \frac{1}{2}$ solves $y'' + 2y = 1$, but $y_1 + y_2 = \frac{3}{2}$ gives $y'' + y = \frac{3}{2} \neq 2$.
- A product like $t^{-1}e^t$, or $\tan t$, $\sec t$, is not of the allowed form, and neither is an equation with a non-constant coefficient such as $y'' + ty = 1$: for a polynomial q of degree n, $q'' + tq$ has degree $n+1$, so it never equals 1.

§3 Homework this week

HW11 (due Nov 16), Nagle Section 4.4: 12, 13, 18, 22, 25, 27, 29, 31 (all on the method of undetermined coefficients, Section 2.1 of this handout). Nagle 4.5 is lectured this week but no posted assignment names it. HW11's Nagle 4.3 exercises are listed in the Week 11 handout. HW11 also includes Lay Chapter 6 Supplementary Exercises 6, 8, 13, 14, 16, 19, 23, 24 (review of Weeks 9–10).

§4 Problem set

No past Math 54 exam is in the course folder; these problems use the same methods with different equations.

P1. Find a particular solution of $y'' - y' - 2y = 4t$.

P2. Find a particular solution of $y'' + 2y' + 5y = 10\cos t$, once with real coefficients and once with the complex exponential.

P3. Find a particular solution of $y'' - 4y = 8e^{2t}$.

P4. Find a particular solution of $y'' + 6y' + 9y = 2e^{-3t}$.

P5. Write the form of a particular solution of $y'' + 4y = f$ (do not solve) for (a) $f = t\sin 2t$; (b) $f = e^{2t}\cos t$; (c) $f = 3t^2e^{2t} + \cos 2t$.

P6. The functions below solve $y'' - y' - 2y = 2t$ and $y'' - y' - 2y = e^t$ respectively:

$$y_1 = -t + \frac{1}{2}, \qquad y_2 = -\frac{1}{2}e^t.$$

Find all solutions of $y'' - y' - 2y = 6t - 4e^t$.

P7. Solve $y'' - y = 2t$ with $y(0) = 0$, $y'(0) = 1$.

P8. Solve $y'' + y = \cos 2t$ with $y(0) = 0$, $y'(0) = 0$, and describe the motion.